

So we want to interface SRAM1 of 8KB and SRAM2 of 4KB to the external memory interface of ATmega162

As a recall,

Port A=> AD0 to AD7

Port C => A8 –A15

PE1=> ALE

PD6=> WR'

PD7=> RD'

For SRAM1:

Address space $2^3 \cdot 2^{10} \Rightarrow$ 13 bit address bus: A0-A12

D0-D7 connected directly to AD0-AD7

A0-A7 connected through latches 74x573 from AD0 to AD7

A8-A12 connected directly from A9 to A12 (PC0-PC4) of port C

OE' connected directly from RD' (PD7)

WE' directly connected to WR' (PD6)

CS connected to the output of AND gate where inputs are A15', A14', A13' which are unused address lines in the address space 0000 H to 1FFF H (0000 0000 0000 0000 B to 0001 1111 1111 1111 H)

Latches:

D0-D7 connected to AD0-AD7 of ATmega162

Q0 to Q7 connected to A0-A7 of SRAM1

E connected to ALE of ATmega162

For SRAM2(4 KB):

Address space $2^2 \cdot 2^{10} \Rightarrow$ 12 bit address bus: A0-A11

D0-D7 connected directly to AD0-AD7

A0-A7 connected through latches 74x573 from AD0 to AD7

A8-A11 connected directly from A8 to A11 (PC0-PC3) of port C=> upper address lines A12 to A15 unused)

OE' connected directly from RD' (PD7)

WE' directly connected to WR' (PD6)

CS connected to the output of AND gate where inputs are A15', A14', A13, A12' which are unused address lines in the address space 2000 H to 2FFF H (0010 0000 0000 0000 B to 0010 1111 1111 1111 B)

The unused address lines are fixed to these values to avoid any address aliases for the same memory address...For e.g. **0010** 0000 0000 0000 B is the same address for SRAM2 as **0011** 0000 0000 0000 B. When we fix the unused address combinations and use them for CS, address aliasing can no longer take place For unstance if we have fixed the unused address bits to 0010, 0011 won't enable SRAM2 and no longer points to the same address as with 0010 combination.

Latches:

D0-D7 connected to AD0-AD7 of ATmega162

Q0 to Q7 connected to A0-A7 of SRAM1

E connected to ALE of ATmega162

